

SciLowCode Softwarepackage Manpage

Manpage (Manual page) for the software package SciLowCode

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1 NAME

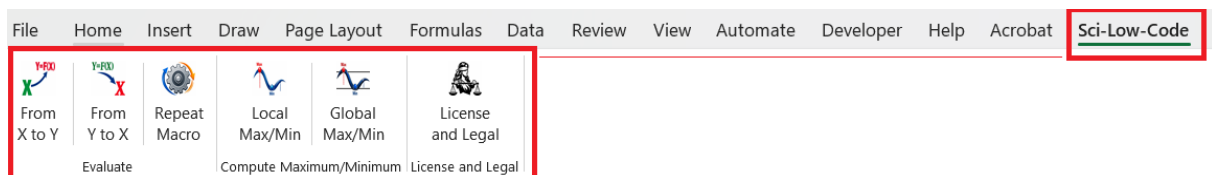
Technical name: **SciLowCode**. This name is the abbreviation of

Science and Low Code Development Library

and aims at fostering Science Computing and Low-Code/No-Code development based on the UI of Excel. SciLowCode is technically an Excel Add-In.

2 SYNOPSIS

SciLowCode must be installed as an Excel Add-In. After activating it appears as a new Entry in Excel's Ribbon Menu:



3 DESCRIPTION

Each button in the menu implements an analytical function. The last button displays the license, legal framework and the activated version of the library. The function in order from left to right:

1. "From X to Y": Mass evaluation of a function $F(X) = Y$ on a panel of data. This mass evaluator is especially useful for evaluation complex functions implemented as spreadsheet calculations.
2. "From Y to X": Mass evaluation of a function $F(X) = Y$ backwards, i.e. given Y find the solution X. This works only for numeric functions since the Newton method https://en.wikipedia.org/wiki/Newton%27s_method is employed. The Newton method is implemented in a DLL (Dynamic Link Library) linked to Excel.

3. "Repeat Macro": A macro is repeated either a specified number of times or as long as a condition holds true. "Infinite loops" are avoided by the input of the upper bound of iterations.
4. "Local Max/Min": The local extremums are computed by this button. A variant of Gradient Methods https://en.wikipedia.org/wiki/Gradient_method is used. This Gradient method is also implemented in a dedicated DLL and linked to Excel.
5. "Global Max/Min": Using gradient methods as in the previous point, the global extremum on a compact interval is computed, guaranteed by for example https://en.wikipedia.org/wiki/Extreme_value_theorem

4 OPTIONS / FLAGS

No special features.

5 USAGE

Please refer to <https://github.com/May-Tinh-Sci-Low-Code/Sci-Low-Code> for downloading the library.

Installing and activating the library is standard in Microsoft Office.

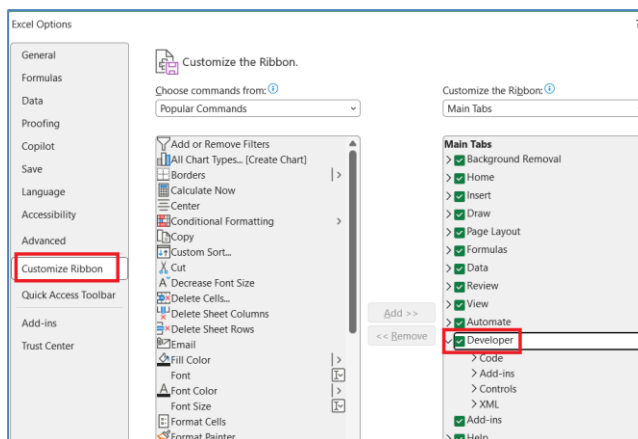
5.1 Activation via Developer menu

This activation needs 2 steps: The first step activates the „Developer“ menu in Excel. In the next we activate SciLowCode via this menu.

5.1.1 Activate „Developer“ menu

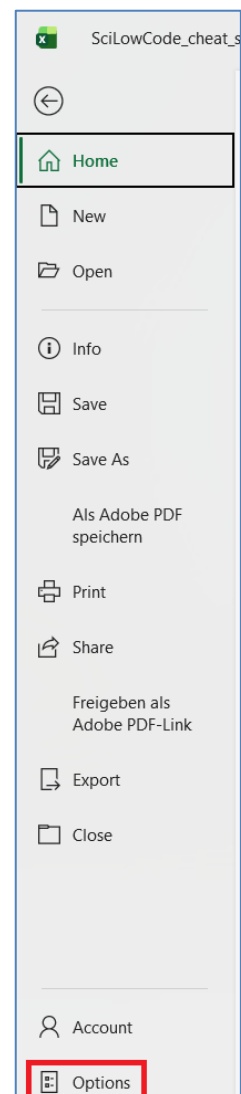
Please follow the next steps:

1. Click „File“ in Excels Ribbon Menu.
2. Choose „Options“ as in the vertical screenshot on this page.
3. In the following pop-up “Excel Options” choose “Customize Ribbon” and then tick “Developer” in the right half of the window:



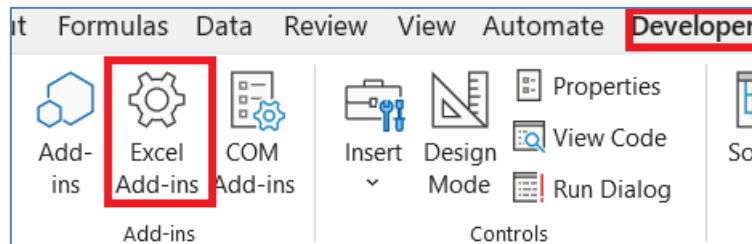
4. Confirm with ok until all pop-ups close.

The „Developer“ menu is active in Excels ribbon.

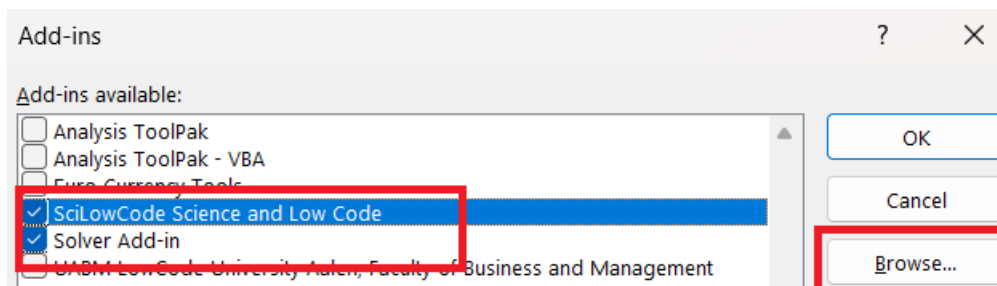


5.1.2 Activate SciLowCode

In menu „Developer“ choose the button „Excel Add-ins“:

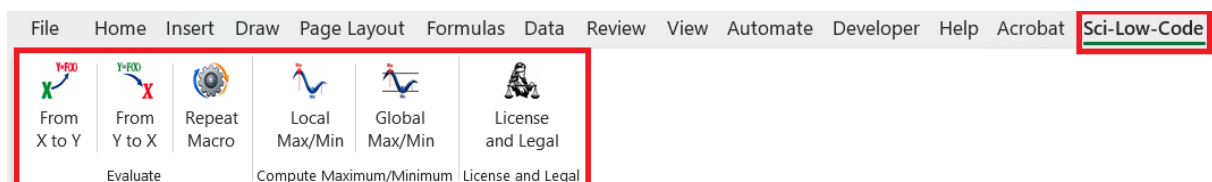


In the Add-ins pop-up choose “Browse...” to import the SciLowCode.xlam. Also, make sure the Add-ins “Solver Add-in” is active too:



Confirm with “OK” button.

Result: The installed Add-In is displayed in Excel’s ribbon menu:

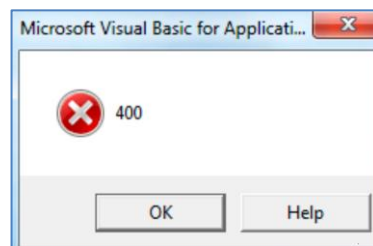


5.2 Activating the Add-in in new Excel versions

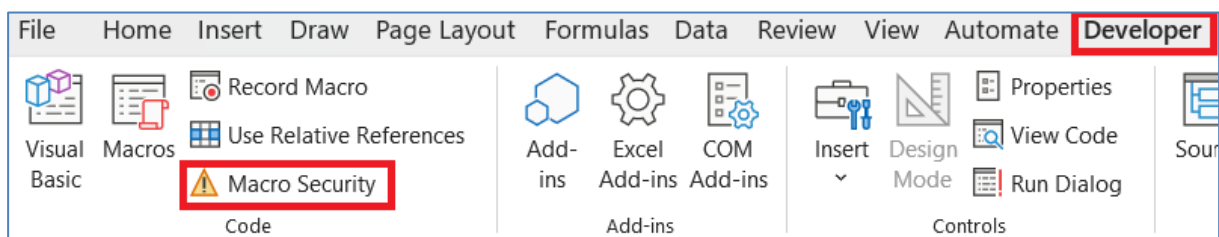
In new Excel versions activating the Add-in is possible by double-clicking on the .xlam file. The "Solver Add-in" must be activated manually.

6 EXIT STATUS

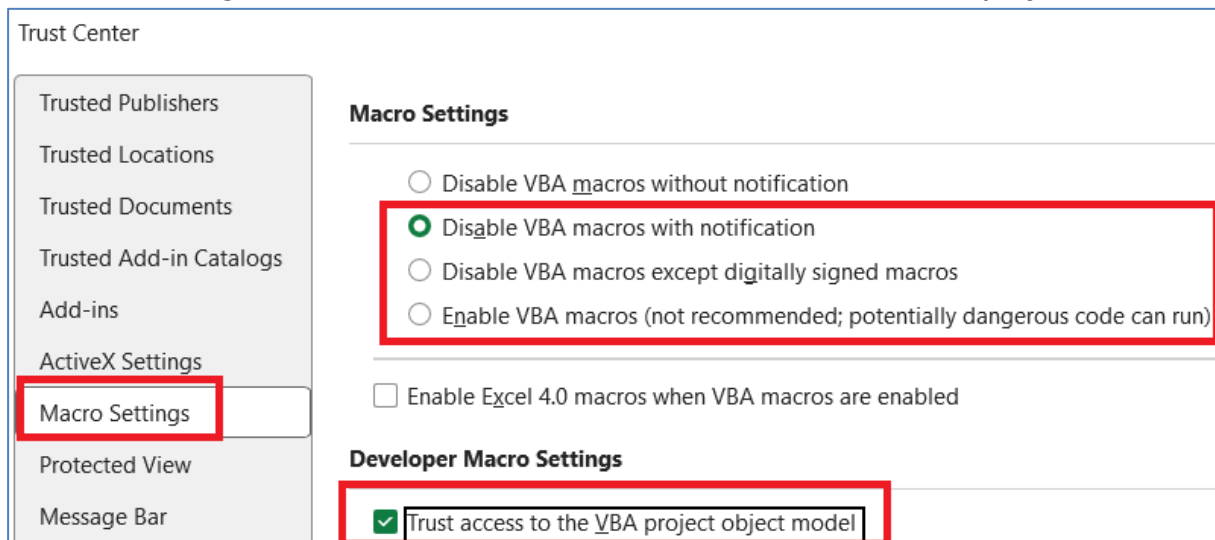
Installing SciLowCode may trigger error 400 as in the screenshot.



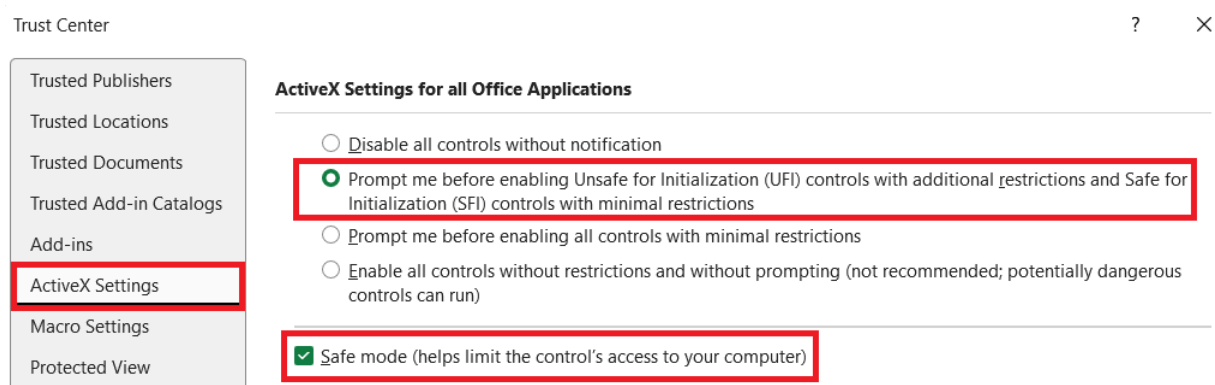
In this case choose menu "Developer" → "Macro Security":



In the next pop-up choose “Macro Settings” and allow for the indicated security in “Macro Settings” as well as “Trust access to the VBA project model”



For operating libraries smoothly check and ensure the following settings are in place:



7 EXAMPLES

Example scenarios will be developed for each functionality separately